

How Machine Learning Is Used for TDS and BMCS

KMAT Truck Cost Model - Simple Technical Documentation

Clear answer

Yes, the project uses machine learning for TDS and BMCS. However, both models currently run in SHADOW mode. They generate predictions for comparison, monitoring and review. They do not control the official cost.

1. Quick Summary

Area	ML Type	Model Output	Current Use	Official Effect
TDS	Regression	Tooling factor from 1.0 to 3.5	Advisory comparison	No current cost authority
BMCS	Classification + probability calibration	Probability that the RFQ mapping is correct	Trust and review advisory	Never a direct cost multiplier

Current control rule

If model input quality is poor, the model fails, or a governance check is not passed, the system uses the deterministic rule value. This is called deterministic fallback.

2. TDS Machine Learning

TDS means Tooling Degradation Score. It estimates how much stress a truck configuration and operation may place on machines and tooling.

Registered model	KMAT_COST_MODEL_DB.CORE_ML.TDS_XGB_REGRESSOR
Model version	V2_SHADOW_1
Feature set	TDS_FEATURES_V2
Algorithm	XGBoost regression
Target	TARGET_TDS_FACTOR

Allowed output range	1.0 to 3.5
Current training data class	Synthetic
Current mode	SHADOW with monitoring

What data does TDS use?

- Truck configuration: engine, cab, wheel and color.
- Operation and machine identity: operation, work center and machine.
- Machine condition: machine age, running hours, vibration, temperature and health score.
- Maintenance condition: days since maintenance, recent maintenance count and recent downtime.
- Workload and complexity: setup time, tool cycles, shift load and configuration complexity.

How TDS is used

Machine + configuration data	XGBoost predicts TDS factor	Range and quality checks	Compare with rule factor	Store and monitor
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1. The model predicts a numeric tooling factor.
2. The system checks that the value is valid and stays between 1.0 and 3.5.
3. The ML value is compared with the deterministic TDS factor.
4. Because the model is in SHADOW mode, the deterministic factor remains the official value.
5. The prediction is saved for monitoring and later validation with real outcomes.

TDS example

Rule TDS = 1.40 and ML TDS = 1.58. The system displays both values. In the current deployment, the official factor remains 1.40. If feature quality fails, the system also returns to 1.40.

3. BMCS Machine Learning

BMCS means BOM Match Confidence Score. It estimates the probability that the system correctly mapped an RFQ into KMAT characteristics such as engine, cab, wheel and color.

Registered model	KMAT_COST_MODEL_DB.CORE_ML.BMCS_CALIBRATED_CLASSIFIER
Model version	V2_SHADOW_2
Scoring adapter	BMCS_PHASE7_ADAPTER_V2
Algorithm	XGBoost classifier
Probability calibration	Isotonic calibration
Target	TARGET_CORRECT_MAPPING_FLAG
Output	Calibrated probability and BMCS score from 0 to 100
Current training data class	Synthetic
Current mode	SHADOW, advisory review only

What data does BMCS use?

- RFQ details: document length, page count, language and OCR quality.
- Document difficulty: complexity, ambiguity, conflicting phrases and unsupported values.
- Extracted configuration: engine, cab, wheel and color.
- Extraction confidence and text evidence for each characteristic.
- Validation evidence: valid values, missing requirements, exact matches, synonym matches and rule checks.
- Existing rule BMCS score and Cortex confidence score.

How BMCS is used

RFQ + extraction evidence	Classifier predicts correct mapping	Isotonic calibration	Advisory trust score	Engineering review gate
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1. The classifier predicts whether the RFQ-to-KMAT mapping is likely to be correct.
2. Isotonic calibration improves the meaning of the probability. For example, a value near 0.90 should represent high confidence more reliably.
3. The probability is converted to a BMCS score from 0 to 100 and an advisory trust band.
4. The ML result is shown beside the official rule/Cortex BMCS result.
5. BMCS is used for trust and review. It never directly increases or decreases cost.

BMCS example

Rule BMCS = 95.70 and calibrated ML probability = 0.9415, which becomes an ML score of 94.15. The ML score is useful evidence, but the official review and trust decision still follows the governed process.

4. Why We Use SHADOW Mode

- Both TDS and BMCS were trained and validated using synthetic data in the current project.
- Synthetic data is suitable for development and demonstration, but it is not enough for automatic production decisions.
- SHADOW mode lets us compare ML predictions with deterministic rules without creating financial risk.
- Real production data and approved performance limits are required before any controlled promotion.

5. Final System Behaviour

Question	TDS	BMCS
Does ML run?	Yes	Yes
Is the prediction stored?	Yes	Yes
Is it monitored?	Yes	Yes
Does it control official cost now?	No	No
What remains official?	Deterministic TDS factor	Governed review/trust result
What happens on failure?	Rule fallback	Governed review fallback

Simple interview answer

We use an XGBoost regression model for TDS to estimate tooling strain, and a calibrated XGBoost classification model for BMCS to estimate whether an RFQ mapping is correct. Both models run inside Snowflake in SHADOW mode. Their predictions are stored, compared and monitored, but deterministic rules and governed review remain official.

Source Basis

This document is based on the project feature-engineering document and the implemented TDS/BMCS model registration, scoring and governance scripts stored in the KMAT project files.